Experiment 6
Exploration of Cloud Service Models (IaaS, PaaS, SaaS) Using a Free-Tier Cloud Platform

Aim

To explore the three main cloud service models, namely Infrastructure as a Service (IaaS), Platform as a Service (PaaS) and Software as a Service (SaaS), using the free tier of a public cloud platform.

Objectives

- To understand what cloud computing is and why organisations use it.
- To create a free-tier account on a public cloud platform.
- To launch a basic virtual machine (IaaS), deploy a small web application (PaaS) and use a ready-made online application (SaaS).
- To compare the three models based on control, cost and maintenance effort.

Requirements

Item	Details
Cloud platform	AWS Free Tier (Google Cloud or Microsoft Azure free accounts can also be used)
Hardware	Computer with at least 4 GB RAM and internet connection
Software	Web browser, SSH client (PuTTY or terminal)
Other	Valid email address and a debit/credit card for account verification (no charge is made if usage stays within the free limit)

Introduction

Cloud computing means using computing resources such as servers, storage, databases and software over the internet instead of owning them. The user pays only for what is used, and the provider takes care of the hardware. Cloud services are mainly offered in three models. They differ in how much work is handled by the provider and how much is left to the customer.

Comparison of Service Models

Feature	IaaS	PaaS	SaaS
What is provided	Virtual servers, storage, network	Platform to build and run applications	Complete ready-to-use software
User manages	Operating system, runtime, application, data	Application and data only	Only own data and settings

Feature	IaaS	PaaS	SaaS
Provider manages	Hardware, virtualisation	Hardware, OS, runtime	Everything
Flexibility	Very high	Medium	Low
Typical user	System administrators	Developers	End users
Examples	Amazon EC2, Google Compute Engine	AWS Elastic Beanstalk, Google App Engine	Gmail, Google Docs, Zoho CRM

Procedure

Part A: Account Setup

1. Open the website of the cloud provider and click on "Create a Free Account".
2. Enter the email address, password and account name, then verify the email.
3. Fill in contact and payment details and complete phone verification.
4. Log in to the management console and note the free-tier usage limits shown on the billing page.

Part B: IaaS - Launching a Virtual Machine

1. Open the compute service (EC2) and click "Launch Instance".
2. Select a free-tier eligible operating system image such as Amazon Linux or Ubuntu.
3. Choose the smallest instance type marked as free-tier eligible (for example t2.micro or t3.micro).
4. Create a key pair, allow SSH and HTTP in the security group, and launch the instance.
5. Connect to the instance using SSH, install a web server (for example Apache) and open the public IP address in a browser to check the default page.
6. Stop and terminate the instance after the experiment to avoid charges.

Part C: PaaS - Deploying a Web Application

1. Open the platform service (Elastic Beanstalk or App Engine) and create a new application.
2. Select the runtime, for example Python or Node.js, and upload the sample application package.
3. Wait for the environment to be created and open the URL given by the platform.
4. Observe that the operating system, web server and scaling are handled automatically.

Part D: SaaS - Using an Online Application

1. Log in to Google Docs or Zoho Writer using a browser.
2. Create a document, share it with another user and edit it at the same time.
3. Observe that no installation, update or server setup is needed.

Observations

Service Model	Service Used	Time Taken	Effort Required from User
IaaS	Virtual machine (EC2)	About 10 minutes	High - server setup and software installation done by user
PaaS	Elastic Beanstalk / App Engine	About 8 minutes	Medium - only code upload needed
SaaS	Google Docs / Zoho Writer	Less than 1 minute	Very low - ready to use

Result

A free-tier cloud account was created and one service from each model was used successfully. It was observed that the control given to the user is highest in IaaS and lowest in SaaS, while the ease of use is lowest in IaaS and highest in SaaS.

Conclusion

The experiment showed that the choice of a cloud service model depends on the need of the user. IaaS suits those who need full control over the system, PaaS suits developers who want to concentrate only on the code, and SaaS suits ordinary users who need ready-made software. Free tiers make it possible for students to learn these services without any cost, provided the resources are stopped after use.